



National Institute of Technology, Tiruchirappalli

Department of Physics-Final Assessment

Date: 22/12/2025

Subject: Physics-I (PHIR11)

Time: 9.30 am -12.30 pm

Branch and Section: ICE-B

Answer all Questions

50 Marks

1. (a) Using thermal equilibrium considerations, derive the expressions relating Einstein's absorption, spontaneous and stimulated transition coefficients A_{21} , B_{12} , and B_{21} . ✓ (5) (5)
(b) If the pulse width of a laser ($\lambda = 10640 \text{ \AA}$) is 20 ms and average output power per pulse is 0.8W, determine how many photons do each pulse contain? ✓ (0) (3)
(c) Differentiate between direct band gap and indirect band gap semiconductors. Explain why direct band gap semiconductors are preferred for semiconductor lasers? ✓ (2) (2)
 2. (a) Explain step-index and graded-index optical fibers with their refractive index profiles, and illustrate the modes of light propagation in each. Also state which type of optical fiber is used in telephone lines. (5) (4.5)
(b) Describe the following terms in optical fiber with neat diagrams: (i) Total internal reflection
(ii) Numerical aperture. ✓ (3) (3)
(c) Consider a fibre having a core of index 1.48, a cladding of index 1.46 and has a core diameter of 30 mm. Prove that all rays making an angle less than 9.43° with the axis will propagate through the fibre. ✓ (2) (1.2)
 3. (a) State the assumptions of Planck's theory of black body radiation and derive Planck's energy density equation for black body radiation. ✓ (5) (5)
(b) Describe Davisson and Germer experiment with a neat diagram. What does it confirm? ✓ (3) (3)
(c) An electron is moving in a one-dimensional infinite well of width $2 \times 10^{-10} \text{ m}$. Find the probability of locating the electron between $x = 0$ and $x = 1 \times 10^{-10} \text{ m}$ in the first excited state? ✓ (2) (2)
 4. (a) What is mean by radioactive decay? State the law of radioactive decay with its mathematical expression of half-life, activity and mean life ✓ (5) (5)
(b) What is your understanding on binding energy of a nucleus? How does the binding energy per nucleon vary with mass numbers for different elements in the periodic table? ✓ (1.5) (3)
(c) What basis elementary particles are classified? Show the classifications of fermions. (0) (2)
 5. (a) Derive the Drude -Lorentz theory of electrical conductivity for metals with a neat diagram. ✓ (0.5) (4)
(b) Show how the electric displacement (D) is related to the electric field (E) and polarization (P) in dielectrics. ✓ (2) (2)
(c) The critical temperature of a superconductor at zero magnetic field is T_c . Determine the temperature at which the critical field becomes half of its value at 0K. ✓ $H = H(0K) \left(1 - \frac{T^2}{T_c^2}\right)$ (2) (2)
(d) What is superconductivity? Explain the Meissner effect with a neat diagram? ✓ (2) (2)
- [Constants: $k_B = 1.38 \times 10^{-23} \text{ J/K}$; $e = 1.602 \times 10^{-19} \text{ C}$; $h = 6.626 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ m/s}$; $1 \text{ eV} = 1.60218 \times 10^{-19} \text{ J}$; $m_e = 9.11 \times 10^{-31} \text{ kg}$]
